

Distributed Optimal Scheduling of LEO Constellations for Ground Based Assets

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and Uncertain Systems



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Overall Framework

2.1. Redstone's Problem

$$\begin{aligned} \min \quad & \mathbf{1}^T \mathbf{G}^2 \mathbf{z} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N \\ & \mathbf{z} \geq \mathbf{E}\mathbf{x} + \mathbf{f} - \mathbf{1}, \quad \mathbf{z} \geq \mathbf{0} \end{aligned}$$

Modeling Operation Cadence Constraint

Modeling Revisit Time Cost Function

2.2 Auxiliary Functions

- Adjust τ for each constraint element
- Adjust G for each cost function element
- Place multiple payloads in constraint object

3. SAT-to-GS

Constraint: SAT Ops Cadence
Cost Function: Each GS Revisit

4. GS-to-SAT

Constraint: GS Ops Cadence
Cost Function: Each SAT Revisit

3.1. Earth Observation SAT Image Capture Scheduling

- Get image data from ground by any available satellite
- Minimize revisit time for each ground operation point

3.2. 6G SAT Data Downlink/Uplink Scheduling (With ISL)

- Uplink/Downlink data packet from GS to any satellite
- Minimize time gap between each transmission
- Assume every satellite is connected each other by ISL

4.1. Distributed Ground Asset Scheduling for SSA

- Get SSA data of satellite by any available ground asset
- Minimize observation gap for each satellite

4.2. Earth Observation SAT Data Download Scheduling

- Download image data to any available ground station
- Evaluate data acquisition performance for each image

REDSTONE Formulation

REvisit-time Determination and Satellite Tasking Optimization Equations

L_1 Problem: Maximize number of contacts
 $\max \mathbf{1}^T \mathbf{x}$
 $\text{s.t. } \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$

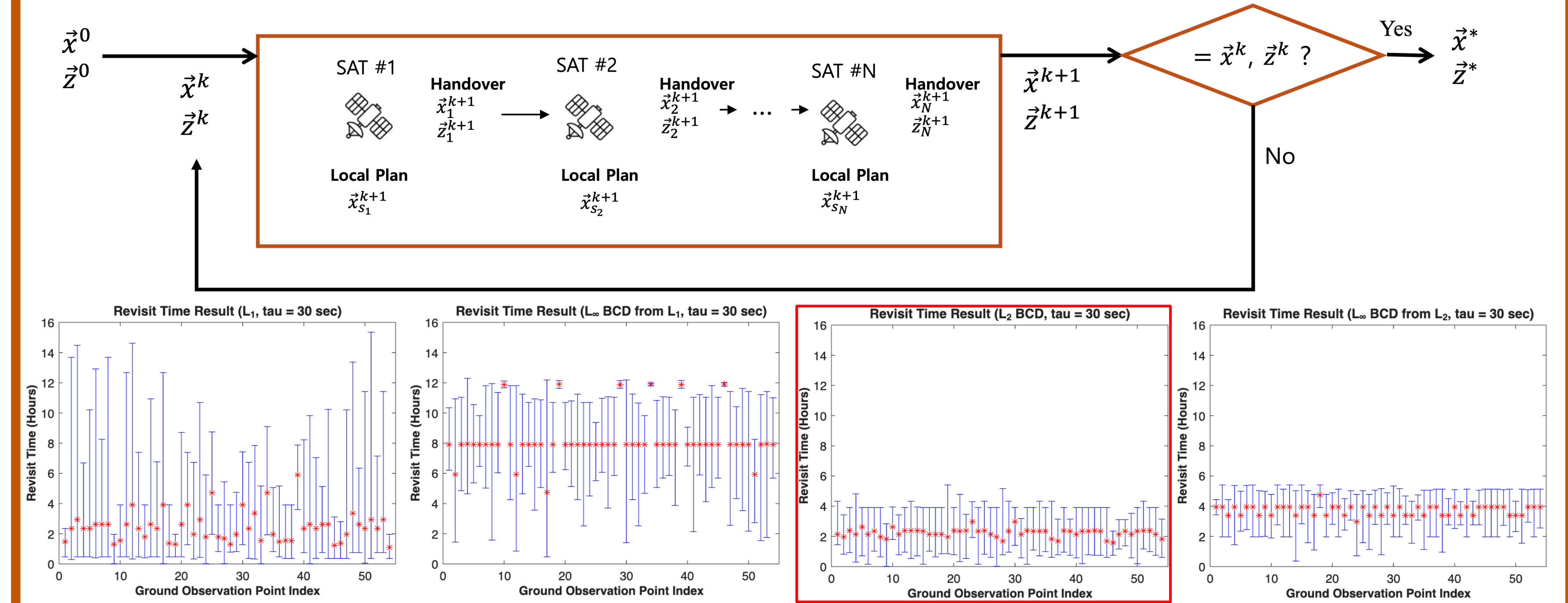
L_2 Problem: Minimize Square-sum Revisit Time
 $\min \mathbf{1}^T \mathbf{G}^2 \mathbf{z}$
 $\text{s.t. } \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$
 $\mathbf{z} \geq \mathbf{E}\mathbf{x} + \mathbf{f} - \mathbf{1}, \quad \mathbf{z} \geq \mathbf{0}$
 $\mathbf{z} \in R^M, M = \sum_{j=1}^q (|G_j| + 2)(|G_j| + 1)$

L_∞ Problem: Minimize Maximum Revisit Time
 $\min R$
 $\text{s.t. } \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$
 $R \mathbf{1} \geq \mathbf{G}(\mathbf{E}\mathbf{x} + \mathbf{f} - \mathbf{1})$

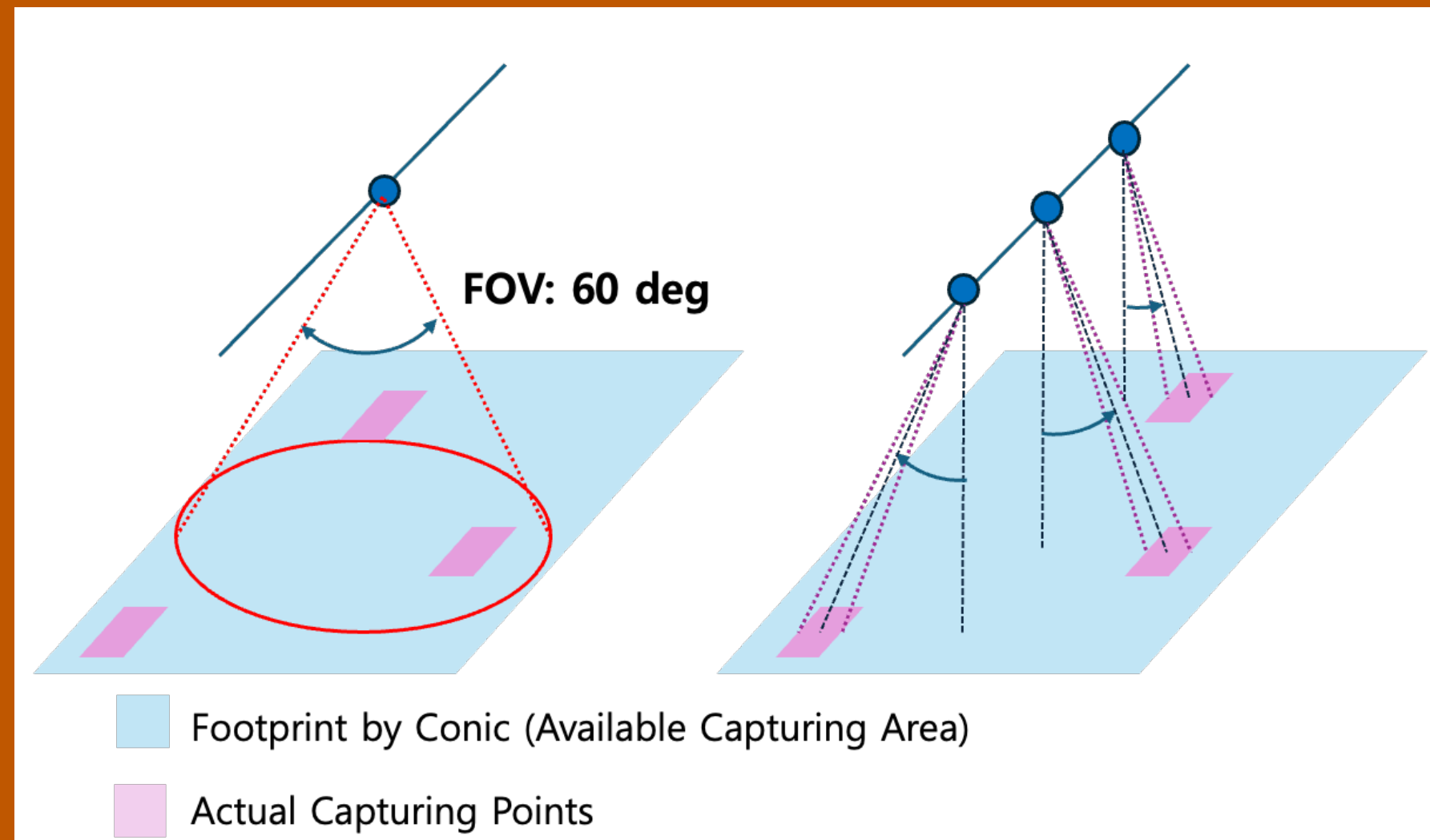
Constants

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} E_{S_1,x}^2 & E_{S_1,t}^1 & E_{S_1}^1 \\ E_{S_2,x}^2 & E_{S_2,t}^1 & E_{S_2}^1 \\ \vdots & \vdots & \vdots \\ E_{S_p,x}^2 & E_{S_p,t}^1 & E_{S_p}^1 \end{bmatrix}, \quad \mathbf{b} = \mathbf{1} + \frac{1}{\tau} \begin{bmatrix} E_{S_1,t}^2 & E_{S_1}^1 \\ E_{S_2,t}^2 & E_{S_2}^1 \\ \vdots & \vdots \\ E_{S_p,t}^2 & E_{S_p}^1 \end{bmatrix} \mathbf{t} \\ \mathbf{E} &= \begin{bmatrix} E_{G_1,x}^2 & E_{G_1,t}^1 & E_{G_1}^1 \\ E_{G_2,x}^2 & E_{G_2,t}^1 & E_{G_2}^1 \\ \vdots & \vdots & \vdots \\ E_{G_q,x}^2 & E_{G_q,t}^1 & E_{G_q}^1 \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} E_{G_1,t}^2 & E_{G_1}^1 \\ E_{G_2,t}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{0}_{N \times 1} \end{bmatrix} \\ \mathbf{G} &= \text{diag} \left(\begin{bmatrix} E_{G_1,t}^2 & E_{G_1}^1 \\ E_{G_2,t}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \mathbf{t} \\ t_{\text{end}} \end{bmatrix} \right) \end{aligned}$$

Sequential Block-Coordinate Descent

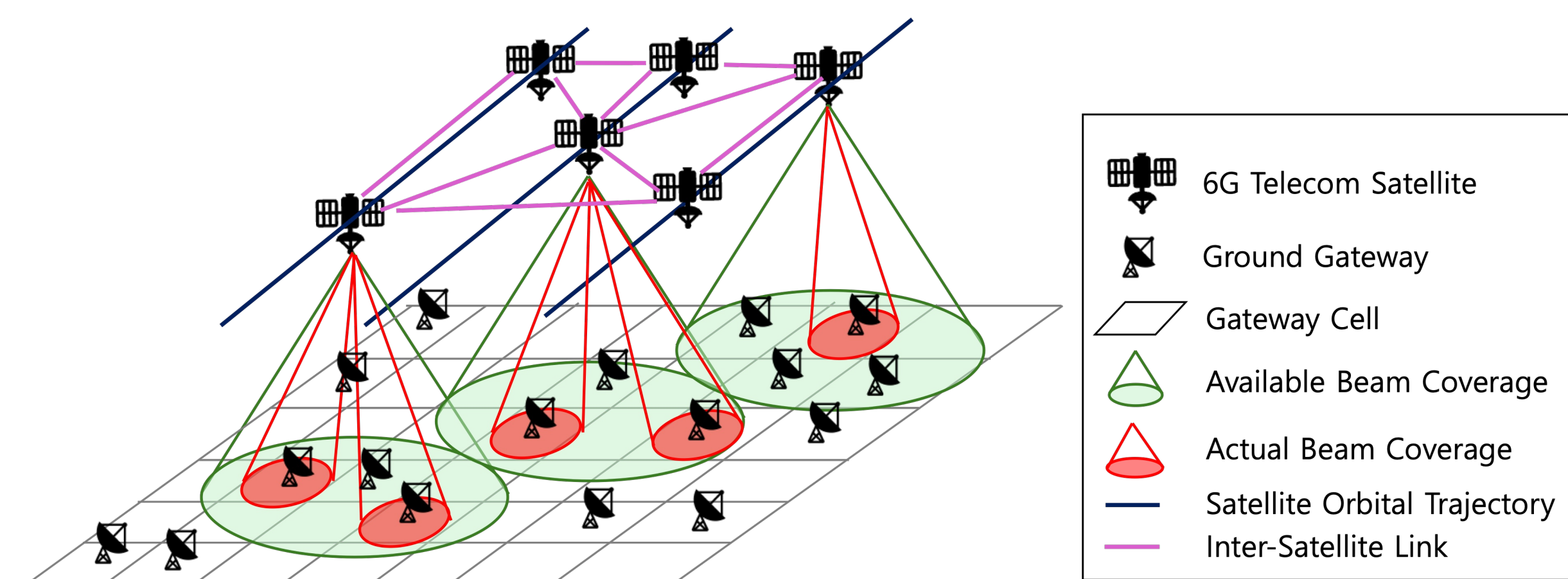


Earth-Observation



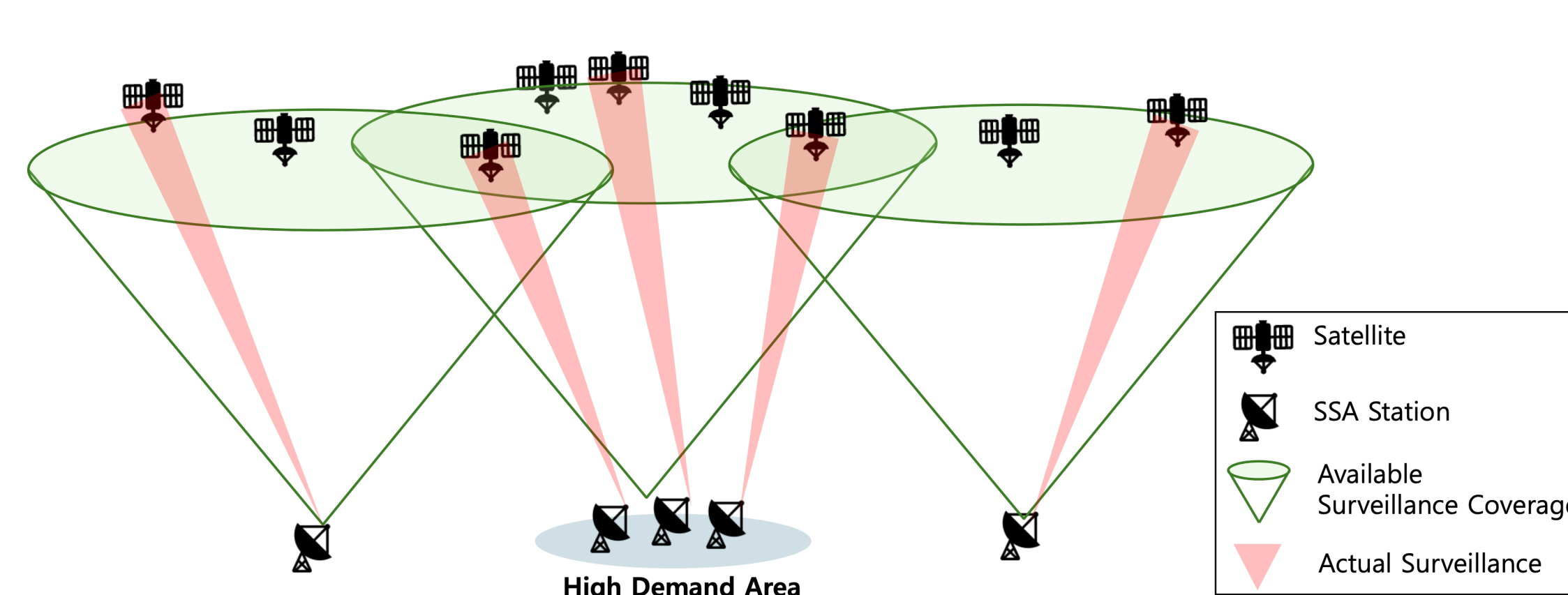
Parameter	Earth Observation Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 7 days, Timestep: 5 seconds
Ground Points	31 globally distributed ground observation points
Orbit: Altitude, Shape	500 km, Circular Orbit
Orbit: Inclination Angle	98°
Payload FOV	60°
Specific Mission	Simple Conic Payload FOV, Slew roll angle for capturing
Constellation Config.	48 SATs: 12 Orbit Planes, 4 SATs/plane
No. of Total Missions	1500 → Actual Duration: 24 hours

6G Telecommunication Network



Parameter	6G Telecom Constellation Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 1 hour, Timestep: 5 seconds
Ground Points	56 hexagonal cell grid points in South Korea
Orbit: Altitude, Shape	1200 km, Circular Orbit
Orbit: Inclination Angle	38°
Payload FOV	60°
Specific Mission	Multiple communications payloads (1, 2, 3, 4) per satellite
Constellation Config.	128 SATs: 16 Orbit Planes, 8 SATs/plane
satellite operation cadence (τ)	20s

Distributed SSA Scheduling



Parameter	Distributed Ground Asset Scheduling Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 1 day, Timestep: 5 seconds
Ground Assets	9 selected ground assets distributed worldwide
Orbit: Altitude, Shape	500 km, Circular Orbit
Orbit: Inclination Angle	98°
Minimum Elevation	57.3°
Specific Mission	Multiple ground assets deployed in a high-demand area (Svalbard)
Constellation Config.	48 SATs: 12 Orbit Planes, 4 SATs/plane

Selected Simulation Results

1. Earth Observation SAT image Capture Scheduling

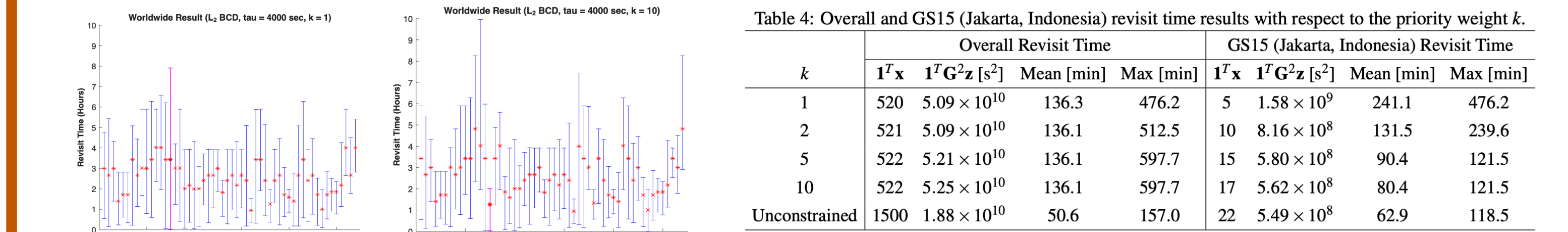


Table 4: Overall and GS15 (Jakarta, Indonesia) revisit time results with respect to the priority weight k .								
k	Overall Revisit Time				GS15 (Jakarta, Indonesia) Revisit Time			
	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s^2]	Mean [min]	Max [min]	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s^2]	Mean [min]	Max [min]
1	520	5.09×10^{10}	136.3	476.2	5	1.58×10^9	241.1	476.2
2	521	5.09×10^{10}	136.1	512.5	10	8.16×10^8	131.5	239.6
5	522	5.21×10^{10}	136.1	597.7	15	5.80×10^8	90.4	121.5
10	522	5.25×10^{10}	136.1	597.7	17	5.62×10^8	80.4	121.5
Unconstrained	1500	1.88×10^{10}	50.6	157.0	22	5.49×10^8	62.9	118.5

2. 6G SAT Data Downlink/Uplink Scheduling (with ISL)

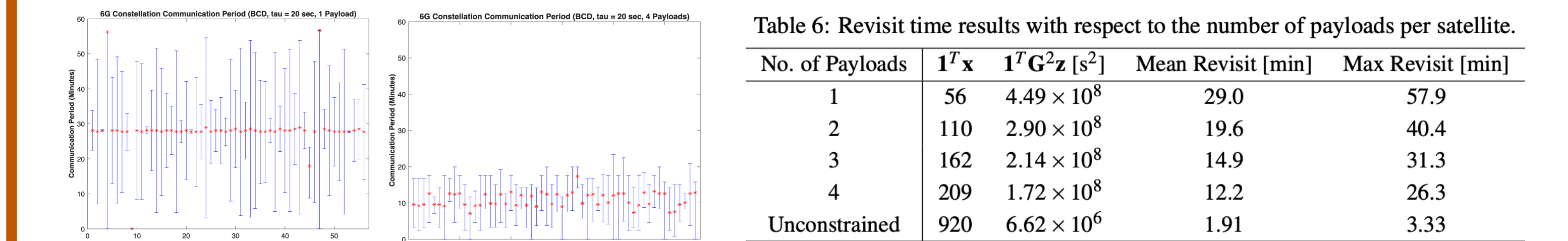


Table 6: Revisit time results with respect to the number of payloads per satellite.				
No. of Payloads	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s^2]	Mean Revisit [min]	Max Revisit [min]
1	56	4.49×10^8	29.0	57.9
2	110	2.90×10^8	19.6	40.4
3	162	2.14×10^8	14.9	31.3
4	209	1.72×10^8	12.2	26.3
Unconstrained	920	6.62×10^6	1.91	3.33

3. Distributed Ground Asset Scheduling for SSA

Table 8: Revisit time results with respect to operation cadence τ for distributed ground asset scheduling.					Table 9: Overall and individual ground-station revisit time results with respect to the number of ground stations.										
τ [s]	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s^2]	Mean Revisit [min]	Max Revisit [min]	Number of GS at Svalbard	Overall Revisit Time				GS Index	Individual GS Revisit Time				
						$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s^2]	Mean [min]	Max [min]		$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s^2]	Mean [min]	Max [min]	
Unconstrained	671	3.86×10^{10}	95.4	427.8	1	275	6.97×10^{10}	213.6	558.3	1	33	2.37×10^9	42.3	71.5	
500	468	4.60×10^{10}	134.0	428.3	2	303	6.43×10^{10}	196.6	558.3	2	33	2.39×10^9	42.3	70.3	
1000	398	5.05×10^{10}	155.0	429.7	3	341	5.99×10^{10}	177.4	556.6	3	33	2.33×10^9	42.3	70.4	
1500	275	6.97×10^{10}	213.6	558.3	4	370	5.83×10^{10}	166.3	556.6	4	34	2.27×10^9	41.1	66.8	
2000	246	7.55×10^{10}	235.5	649.5	—	—	—	—	—	—	Single GS	38	2.05×10^9	36.9	66.9

4. Earth Observation Imaging-to-Download Sequence

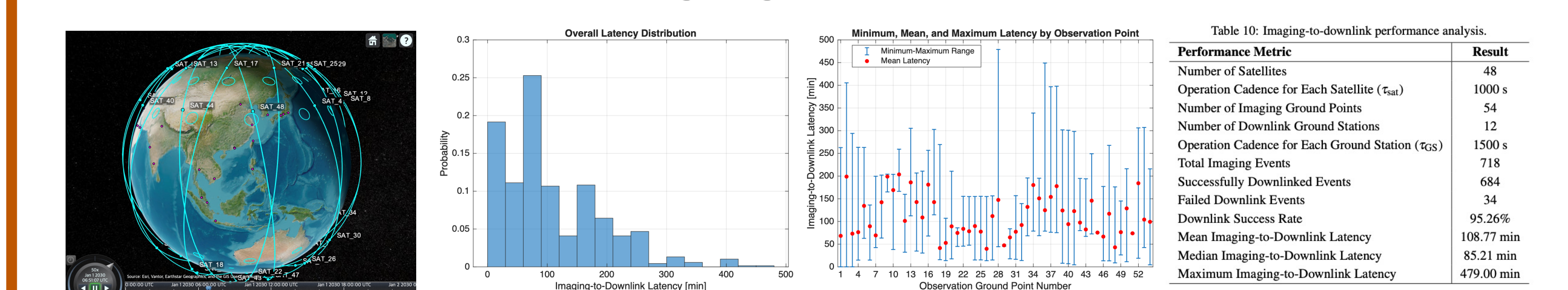


Table 10: Imaging-to-downlink performance analysis.	
Performance Metric	Result
Number of Satellites	48
Operation Cadence for Each Satellite (τ_{sat})	1000 s
Number of Imaging Ground Points	54
Number of Downlink Ground Stations	12
Operation Cadence for Each Ground Station (τ_{gs})	1500 s
Total Imaging Events	718
Successfully Downlinked Events	684
Failed Downlink Events	34
Downlink Success Rate	95.26%
Mean Imaging-to-Downlink Latency	108.77 min
Median Imaging-to-Downlink Latency	85.51 min
Maximum Imaging-to-Downlink Latency	479.00 min